The University of Sydney

MOOC Introduction to Calculus

Notes for 'The quadratic formula'

Important Ideas and Useful Facts:

(i) Solving a quadratic equation by inspection: In simple cases, one is able to solve a quadratic equation by factorising the quadratic by inspection. This technique relies on the fact that

$$(x+a)(x+b) \;=\; x^2+(a+b)x+ab\,.$$

This method works best when the coefficient of x^2 is 1. One looks for numbers that add up to give the coefficient of x and multiply to give the constant term. For example, consider the quadratic equation

$$x^2+4x-12 \;=\; 0\,.$$

Thinking about the integer factorisations of -12, one can see that $-12=(+6)\times(-2)$ and $(+6)+(-2)=4$, so the above formula will work with $a=6$ and $b=-2$. Hence the equation becomes

$$(x+6)(x-2) \;=\; 0\,,$$

so that either $x+6=0$ or $x-2=0$, yielding solutions $x=-6$ and $x=2$.

(ii) The quadratic formula: A *quadratic equation* has the form $ax^2+bx+c=0$ where a, b, c are constants, $a\neq 0$, and has solutions (also called *roots of the quadratic*) given by the *quadratic formula*:

$$x \;=\; \frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

We call $\Delta=b^2-4ac$ the *discriminant*. If $\Delta>0$ then there are exactly two real roots. If $\Delta=0$ then there is exactly one real root.

If $\Delta<0$ then there are no real roots (though exactly two *complex roots* exist in the complex number system $\mathbb{C}$, though treatments involving complex numbers go beyond the scope of this course).

For example, if $a=1$, $b=4$ and $c=-12$ then we are solving the quadratic equation

$$x^2+4x-12 \;=\; 0\,,$$

that we solved earlier, by the factorisation method. If we apply the quadratic formula, we get

$$x \;=\; \frac{-4\pm\sqrt{4^2-(4\times1\times(-12))}}{2\times1} \;=\; \frac{-4\pm\sqrt{64}}{2} \;==\; \frac{-4\pm 8}{2} \;=\; -2\pm 4\,,$$

yielding solutions $x=2$ and $x=-6$, as before.

Examples and derivations:

1. Consider the quadratic equation

$$x^2 - x - 1 \ = \ 0 \ .$$

It is not at all obvious that we can find the solutions by inspection using the factorisation method, so we rely on the quadratic formula where $a = 1$, $b = -1$ and $c = -1$. This gives

$$x \ = \ \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \ = \ \frac{-(-1) \pm \sqrt{(-1)^2 + 4}}{2} \ = \ \frac{1 \pm \sqrt{5}}{2} \ .$$

The positive solution $\frac{1+\sqrt{5}}{2}$ is a famous number in mathematics, known as the *golden ratio*, and is closely connected to the *Fibonacci sequence* and analysis of the complexity of the *Euclidean algorithm*, which you might meet in more advanced courses in mathematics.

2. Consider the quadratic equation

$$16x^2 - 56x + 49 \ = \ 0 \ .$$

It is not clear that an attempt to find solutions by inspection using the factorisation method is likely to bear fruit, so we apply the quadratic formula, where $a = 16$, $b = -56$ and $c = 49$. This gives

$$x \ = \ \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \ = \ \frac{56 \pm \sqrt{(56)^2 - 4(16)(49)}}{32} \ = \ \frac{56 \pm 0}{32} \ = \ \frac{7}{4} \ .$$

In fact, there is only one solution (though we did not know that in advance), because the discriminant $b^2 - 4ac$ turns out to be zero, and the quadratic above factorises as a perfect square:

$$(4x - 7)^2 \ = \ 0 \ .$$

If we had noticed this in the first place, then we would have deduced that $4x - 7 = 0$, so that $x = \frac{7}{4}$.

3. Consider the quadratic equation

$$x^2 - x + 1 \ = \ 0 \ .$$

This looks like a slight modification of the first example (of which the golden ratio was a solution). However, the quadratic formula tells us that if x is a solution then

$$x \ = \ \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \ = \ \frac{1 \pm \sqrt{(-1)^2 - 4}}{2} \ = \ \frac{1 \pm \sqrt{-3}}{2} \ ,$$

which is impossible, working over the real number system $\mathbb{R}$, since we are unable to form the square root of -3. In this case, the discriminant $b^2 - 4ac$ turns out to be negative, so there are no real solutions.

4. We carefully derive the quadratic formula using the method of completing the square. Suppose that

$$ax^2 + bx + c \;=\; 0\;,$$

where a, b and c are constants such that a is nonzero. It is convenient for the coefficient of x^2 to be 1, so we divide through by a to get

$$x^2 + \frac{bx}{a} + \frac{c}{a} \;=\; 0\;,$$

and then take $\frac{c}{a}$ away from both sides:

$$x^2 + \frac{bx}{a} \;=\; -\frac{c}{a}\;.$$

We wish to form a perfect square on the left-hand side, so add the square of half of the coefficient of x, that is, we add

$$\left(\frac{b}{2a}\right)^2 \;=\; \frac{b^2}{4a^2}$$

to both sides, to get

$$x^2 + \frac{bx}{a} + \frac{b^2}{4a^2} \;=\; -\frac{c}{a} + \frac{b^2}{4a^2} \;=\; \frac{b^2 - 4ac}{4a^2}\;.$$

The left-hand side becomes a perfect square, so that

$$\left(x + \frac{b}{2a}\right)^2 \;=\; \frac{b^2 - 4ac}{4a^2}\;.$$

Note that $4a^2$ is positive, and the right-hand side has to be nonnegative (as the left-hand side is a perfect square). This forces $b^2 - 4ac$ to be nonnegative, that is

$$b^2 - 4ac \;\geq\; 0\;.$$

(If this fails, that is, if $b^2 - 4ac < 0$, then this sequence of steps proves that the original quadratic equation has no solution.)

It remains to unravel the equation involving x using algebraic manipulation. We first take positive and negative square roots of both sides, so that

$$x + \frac{b}{2a} \;=\; \pm\sqrt{\frac{b^2 - 4ac}{4a^2}} \;=\; \frac{\pm\sqrt{b^2 - 4ac}}{\sqrt{4a^2}} \;=\; \frac{\pm\sqrt{b^2 - 4ac}}{2a}\;,$$

noting that $\sqrt{4a^2} = 2a$ if $a > 0$, and $\sqrt{4a^2} = -2a$ if $a < 0$, in which case the minus sign may be absorbed in the $\pm$ in the numerator. It remains to take $\frac{b}{2a}$ away from both sides, to get

$$x \;=\; \frac{\pm\sqrt{b^2 - 4ac}}{2a} - \frac{b}{2a} \;=\; \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\;,$$

which completes the derivation of the quadratic formula.

5. Sometimes, equations involving algebraic fractions can be reduced to quadratic equations that are easily solved. For example, consider the equation

$$\frac{2}{x+1} - \frac{2}{x+2} \;=\; 7\,.$$

Note that this equation only makes sense for $x \neq -1$ and $x \neq -2$. We form a common denominator on the left-hand side to get

$$\frac{2(x+2)}{(x+1)(x+2)} - \frac{2(x+1)}{(x+2)(x+1)} \;=\; \frac{2x+4-2x-2}{(x+1)(x+2)} \;=\; 7\,,$$

which becomes

$$\frac{2}{(x+1)(x+2)} \;=\; 7\,.$$

Multiplying both sides by $(x+1)(x+2)$ gives

$$2 = 7(x+1)(x+2) \;=\; 7x^2 + 21x + 14\,,$$

which rearranges to yield the quadratic equation

$$7x^2 + 21x + 12 \;=\; 0\,.$$

Using the quadratic formula, we find

$$x \;=\; \frac{-21 \pm \sqrt{105}}{14}\,.$$